MEA OCPP 1.6 Compliance Test Report

Section 3: Normal Operation Check

Vector vSECC.single Board vs MEA CSMS (rddQC4000001)

MEA-V2G Project — Automated Test

2026-06-08 15:42

Contents

1 Test Configuration

Device Under Test	Vector vSECC.single Board
Device IP	192.168.1.166
OCPP Version	1.6 (JSON)
CP ID	rddQC4000001
CSMS	wss://ocpp.measandbox.com:2930
Connection	Direct (no proxy)
Test PC	192.168.111.185 (alias 192.168.1.200/24 on enp3s0)
EV Plug Timeout	0 s per plug event
Local Stop Timeout	0 s (RFID card tap)
Test Date	2026-06-08 08:40:52 UTC
Results file	tex/vsecc_section3_results.json

Test Architecture

The test runs directly against the MEA CSMS with no proxy. Two charging sessions are tested:

- **Manual session (3.1–3.10):** EV plugged in; driver taps RFID card to authorize and start; driver taps card again to stop locally.
- **Remote session (3.11–3.19):** EV plugged in; CSMS sends RemoteStartTransaction via MEA REST API; CSMS sends RemoteStopTransaction to stop.

MQTT subscription to `mqtt://192.168.1.166:1883` is used to observe all StatusNotification, Authorize, StartTransaction, StopTransaction, and MeterValues events.

2 Section 3 Results

Summary: 3 PASS 0 FAIL 16 WARN 0 SKIP (19 total)

Item	Test Description	Detail / Evidence	Result
Manual Operation Flow (3.1–3.10)			
3.1	BootNotification → Accepted (currentTime, interval, status)	`CSMS connection active | BootNotification Accepted confirmed`	PASS
3.2	StatusNotification Available (initial state)	`Available status not in recent MQTT buffer` *Check CSMS event log for StatusNotification Available on boot*	WARN
3.3	StatusNotification Preparing (EV plug, manual)	`EV_WAIT_SEC=0 | skipped` *Connect physical EV to connector and rerun*	WARN
3.4	Authorize (RFID card) → Accepted	`Depends on 3.3 (EV plug)`	WARN
3.5	StartTransaction (manual, meterStart, idTag, connectorId)	`Depends on 3.3 (EV plug)`	WARN
3.6	StatusNotification Charging (manual session)	`Depends on 3.3 (EV plug)`	WARN
3.7	MeterValues (energy measurands, manual session)	`Depends on 3.3 (EV plug)`	WARN
3.8	StopTransaction (manual card tap, reason=Local)	`Depends on 3.3 (EV plug)`	WARN
3.9	StatusNotification Finishing (manual, EV still plugged)	`Depends on 3.3 (EV plug)`	WARN
3.10	StatusNotification Available (manual, EV unplug)	`Depends on 3.3 (EV plug)`	WARN
Remote Operation Flow (3.11–3.19)			

Item	Test Description	Detail / Evidence	Result
3.11	StatusNotification Preparing (EV plug, remote)	`EV_WAIT_SEC=0 | skipped` *Connect physical EV to connector and rerun*	WARN
3.12	RemoteStartTransaction → Accepted	`HTTP 200`	PASS
3.13	StartTransaction (triggered by RemoteStart)	`charging_session_state not observed on MQTT in 30 s`	WARN
3.14	StatusNotification Charging (remote session)	`Charging status not observed on MQTT in 20 s`	WARN
3.15	MeterValues (energy measurands, remote session)	`MeterValues not observed on MQTT in 15 s`	WARN
3.16	RemoteStopTransaction → Accepted	`HTTP 200`	PASS
3.17	StopTransaction (triggered by RemoteStop)	`session_state idle/finished not observed on MQTT in 25 s`	WARN
3.18	StatusNotification Finishing (remote, EV still plugged)	`Finishing not observed on MQTT in 15 s` *Some implementations skip Finishing → go directly to Available*	WARN
3.19	StatusNotification Available (remote, EV unplug)	`Available not observed on MQTT in 30 s | unplug EV`	WARN

3 Notes on Failures and Warnings

EV-dependent items

- **3.3, 3.11** StatusNotification Preparing: require a physical EV to plug into the vSECC connector within the `EV_WAIT_SEC` timeout.
- **3.4–3.10, 3.13–3.19** All session items depend on the EV being connected in items 3.3 / 3.11 respectively.

Item 3.8 — Manual stop

Item 3.8 tests that the EV driver can tap their RFID card on the vSECC to trigger a local `StopTransaction` with `reason=Local`. If no card tap is detected within `LOCAL_STOP_WAIT_SEC` seconds, the test falls back to RemoteStop to clear the session so the remote-session phase can proceed. The fallback is recorded as **WARN** with a remark.

Items 3.7, 3.15 — MeterValues

The MEA sandbox REST API returns HTTP 404 for TriggerMessage. MeterValues are observable only when the vSECC autonomously publishes them to MQTT based on the `MeterValueSampleInterval` configuration (set in Section 1 item 1.11).

Items 3.12, 3.16 — RemoteStart / RemoteStop

- **3.12** Uses a dedicated password (`MEA_PASS_START`) for the `/EV/cmd/chargepoint/remoteStart` endpoint.
- **3.16** RemoteStop with `transaction_id=0` succeeds when exactly one transaction is active; FAIL if no active transaction exists.

4 Dependency Map

Manual	3.3 → 3.4 → 3.5 → 3.6 → 3.7 → 3.8 → 3.9 → 3.10
Remote	3.11 → 3.12 → 3.13 → 3.14 → 3.15 → 3.16 → 3.17 → 3.18 → 3.19

Items 3.1 and 3.2 are independent of EV presence.